

The strong-coupling gap of the Kogut–Susskind Hamiltonian is uniform in the volume

Statement T2 of the obligations map holds in Hamiltonian form: for every compact connected gauge group G and every lattice size, the Kogut–Susskind Hamiltonian at coupling $g \geq g_0(G)$ has a non-degenerate ground state and a gap

$$\Delta_{a,L} \geq \gamma \frac{g^2}{2} C_2(R_{\min}) \frac{\hbar c}{a},$$

with $\gamma > 0$ independent of the lattice size, and the physical (gauge-invariant) gap is at least as large. The proof is a hypothesis check against Yarotsky’s Theorem 1 on relatively bounded perturbations of classical lattice systems (Commun. Math. Phys. 261 (2006) 799, Sections 1 and Theorem 1 read at passage level): the electric term is a classical Hamiltonian in the Peter–Weyl partition with a product ground state and unit local gap after normalization, and the magnetic term is a bounded, translation-invariant, finite-range perturbation whose normalized size is $\beta(g) = 48N^2/[g^4(N^2 - 1)]$ for $SU(N)$, so the theorem’s smallness condition $\beta \leq \beta_*(3, \Lambda_0)$ is the strong-coupling condition $g^4 \geq 48N^2/[(N^2 - 1)\beta_*]$. The same theorem gives the infinite-volume ground state and exponential clustering at these couplings. The constants β_* and γ are existential in Yarotsky’s statement, so g_0 is a well-defined but not numerically explicit threshold. The result does not distinguish abelian from non-abelian groups; that distinction is T2’, and the rest of the problem is the descent of g_0 to zero along the scaling curve. No claim is promoted.

1. The Hamiltonian in Yarotsky’s normalization

Notation as in the obligations map §1: periodic cubic lattice $\Lambda = (\mathbb{Z}/N_s\mathbb{Z})^3$, links $\mathcal{E}$, plaquettes $\mathcal{P}$, $\mathcal{H} = L^2(G^\mathcal{E})$, and

$$H = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \operatorname{Re} \operatorname{tr} U_p) \right].$$

Let $C_2 \equiv C_2(R_{\min})$ be the smallest nonzero Casimir eigenvalue of $-\Delta$ on $L^2(G)$; for $SU(N)$ with $\operatorname{tr} T^a T^b = \frac{1}{2} \delta^{ab}$, $C_2 = (N^2 - 1)/(2N)$. Divide by the unit $\varepsilon_0 = (g^2/2)C_2 \hbar c/a$:

$$\frac{H}{\varepsilon_0} = \sum_{x \in \Lambda} h_x + \sum_{x \in \Lambda} \phi_x, \quad h_x = \frac{1}{C_2} \sum_{i=1}^3 (-\Delta_{(x, x+e_i)}), \quad \phi_x = \frac{4}{g^4 C_2} \sum_{i < j} (N - \operatorname{Re} \operatorname{tr} U_{p(x, i, j)}),$$

where $p(x, i, j)$ is the plaquette with corner x spanned by e_i, e_j . Every link belongs to exactly one h_x and every plaquette to exactly one ϕ_x , so the two sums reproduce H/ε_0 .

2. Hypothesis check

Yarotsky’s setting (Section 1 of the paper): a lattice $\mathbb{Z}^\nu$, site spaces $\mathcal{H}_x$ possibly infinite-dimensional with a preferred vector Ω_x , translation-invariant interactions of finite range Λ_0 , a classical part $H_0 = \sum_x h_x$ where h_x is a function of a product partition of unity containing the projection onto Ω_x and satisfies $h_x \Omega = 0$, $h_x \geq 1$ on the orthogonal complement of the product vector, and a perturbation $\sum_x \phi_x$ with each ϕ_x a symmetric form on $\mathcal{H}_{\Lambda_0+x}$ obeying $|\phi_x(v, v)| \leq \alpha \|h_x^{1/2} v\|^2 + \beta \|v\|^2$ with $\alpha < 1$. Theorem 1 there: there are $\alpha_*, \beta_* > 0$ depending only on ν and Λ_0 such that if $\alpha \leq \alpha_*$ and $\beta \leq \beta_*$, then for periodic cubic volumes H_Λ has a non-degenerate ground state with a gap $\gamma > 0$ independent of Λ , the ground states have a weak* thermodynamic limit, and that limit has exponentially decaying correlations.

1. *Lattice and site spaces.* $\nu = 3$; $\mathcal{H}_x = L^2(G)^{\otimes 3}$, the three links leaving x ; Ω_x the constant function. Then $\mathcal{H}_\Lambda = \bigotimes_x \mathcal{H}_x = L^2(G^\mathcal{E})$ and $\Omega_{\Lambda,0}$ is the constant function on $G^\mathcal{E}$.

2. *Partition and classicality.* On each link, Peter–Weyl gives the orthogonal partition of unity by isotypic projections P_R , R ranging over irreducible representations, with P_{triv} the projection onto constants. $-\Delta_\ell = \sum_R C_2(R)P_R$ is a function of this partition, so h_x is a function of the product partition on $\mathcal{H}_x$, and h_x is self-adjoint and unbounded, as allowed.
3. *Local ground state and gap.* $h_x \Omega_x = 0$. On the orthogonal complement of Ω_x in $\mathcal{H}_x$ at least one of the three links carries a nontrivial representation, so $h_x \geq C_2(R_{\min})/C_2 = 1$ there: condition (1) of the paper holds with the range $\Lambda_0 = \{0\}$ for h_x , which may be enlarged to the common range below.
4. *Perturbation.* ϕ_x is multiplication by a continuous function on the twelve links of the three plaquettes at x , whose sites lie in $\Lambda_0 + x$ with $\Lambda_0 = \{0, 1\}^3$; it is symmetric, bounded, and $0 \leq N - \text{Re tr } U_p \leq 2N$ gives

$$|\phi_x(v, v)| \leq \beta \|v\|^2, \quad \beta = \frac{4}{g^4 C_2} \cdot 3 \cdot 2N = \frac{24N}{g^4 C_2} = \frac{48N^2}{g^4(N^2 - 1)} \quad (SU(N)),$$

so condition (2) holds with $\alpha = 0$ and this β .

5. *Translation invariance and periodicity.* h_x and ϕ_x are translates of h_0 and ϕ_0 , and the lattice is a periodic cube, the case in which the paper states Theorem 1.

Theorem (T2, Hamiltonian form). Let $\beta_* = \beta_*(3, \{0, 1\}^3)$ and γ be the constants of Yarotsky’s Theorem 1 for this range. For every g with $48N^2/[g^4(N^2 - 1)] \leq \beta_*$, that is $g \geq g_0(N) = (48N^2/[(N^2 - 1)\beta_*])^{1/4}$, and every $N_s \geq 2$: H has a non-degenerate ground state, and

$$\Delta_{a,L} \geq \gamma \varepsilon_0 = \gamma \frac{g^2}{2} \frac{N^2 - 1}{2N} \frac{\hbar c}{a}.$$

The physical gap obeys $\Delta_{a,L}^{\text{phys}} \geq \Delta_{a,L}$ by Theorem 1(3) of the obligations map. The ground states have a thermodynamic limit ω with exponentially decaying correlations of local observables.

Proof. Items 1–5 are the hypotheses of Yarotsky’s Theorem 1 with $\alpha = 0 \leq \alpha_*$ and $\beta \leq \beta_*$; its conclusions 1–3 are the statements, after multiplying the normalized gap γ by ε_0 . The inheritance by the physical sector is the positivity argument already written for T1. $\square$

For a general compact connected G the same holds with $C_2 = C_2(R_{\min})$ and the bound $N - \text{Re tr } U_p \leq 2 \dim$ in place of $2N$; the threshold is $g_0^4 = 24 \dim / (C_2 \beta_*)$.

3. What the theorem does and does not give

- *Uniformity.* γ depends only on $\nu = 3$ and $\Lambda_0 = \{0, 1\}^3$, so the bound holds for every lattice size at once. The extensive Kato bound of the obligations map §3 is replaced by a local one, which is exactly what the cluster structure of the perturbation provides.
- *Large N .* $\beta \rightarrow 48/g^4$ as $N \rightarrow \infty$, so $g_0(N)$ is bounded in N : the strong-coupling gap is uniform in the rank at fixed bare g in this convention. In the ’t Hooft normalization $\lambda = g^2 N$ the threshold reads $\lambda \geq g_0^2 N$, growing with N .
- *Scale.* The gap is measured in $\hbar c/a$ and is proportional to g^2 : a supplied cutoff scale, as the dimensional census of G07 §4 requires at fixed g . Nothing here survives $a \rightarrow 0$ at fixed g , and T3 asks for the behaviour along $g(a) \rightarrow 0$, outside the theorem’s range.
- *Abelian case.* $U(1)$ satisfies every hypothesis with $C_2(R_{\min}) = 1$ (charge one) and $\dim = 1$, so it too is gapped uniformly in volume at strong coupling. The theorem is blind to the property that distinguishes the groups; that property enters only in T2’, where the abelian gap closes (Guth; Fröhlich–Spencer) and the non-abelian one is conjectured to persist.
- *Explicitness.* β_* and γ exist by the theorem and are not given as numbers in its statement. Extracting them from the proof (Section 2 of the paper, a cluster expansion for the ground-state projection) would give a numerical $g_0(N)$; the descent problem does not depend on that number, only on the structure of the expansion.

- *Relation to the small-volume end.* Yarotsky’s expansion is in the electric basis, where the ground state at $g = \infty$ is the constant function. At weak coupling the ground state concentrates near $U_p = 1$ and the good variables are the constant modes of G07 §3.3, with gap $\delta_1 g^{2/3} \hbar c / L$ in the continuum small box. The two expansions use different bases and different small parameters (g^{-4} and $g^{2/3}$); T2’ is the statement that the gap does not close between them.

4. Consequence for STATE

T2 is closed in Hamiltonian form as a corollary of an established theorem, with the threshold $g_0(N) = (48N^2 / [(N^2 - 1)\beta_*])^{1/4}$. The obligations table moves T2 to “established”; the open items are T2’, T3 and T4. The next theorem-sized target on the lattice route is a weak-coupling counterpart: a variational upper bound and, if possible, a lower bound on the Kogut–Susskind gap at small g on a lattice whose side is small against $\hbar c / \Lambda$, using the constant-mode ground state of C133 as the trial state, to connect the lattice gap $\Delta_{a,L}$ with the continuum small-volume corner and to locate the regime where neither expansion applies.